Using Gaussian distribution to construct fitness functions in genetic programming for multiclass object classification

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Abstract

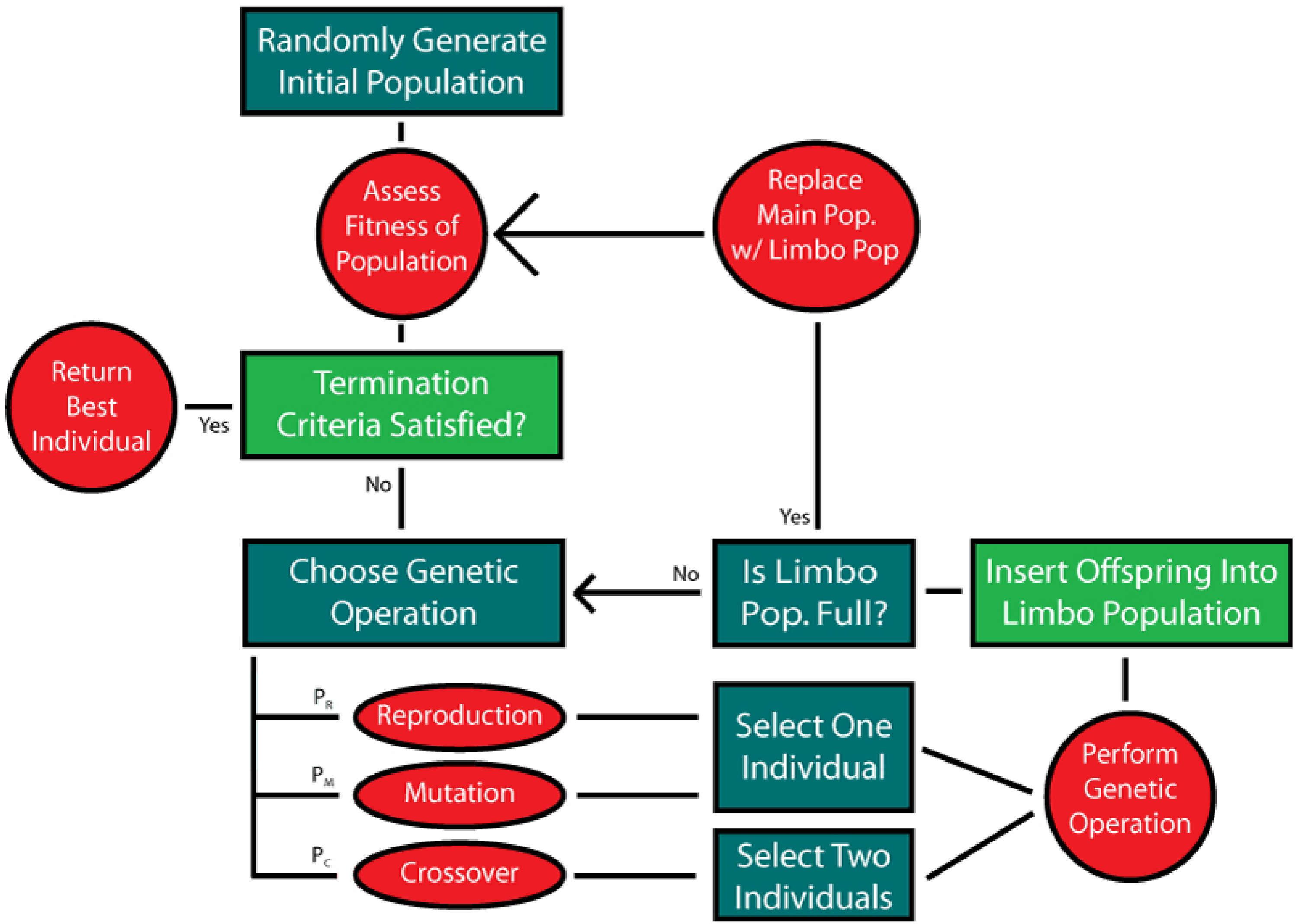
This paper describes a new approach to the use of Gaussian distribution in genetic programming (GP) for multiclass object classification problems. Instead of using predefined multiple thresholds to form different regions in the program output space for different classes, this approach uses probabilities of different classes, derived from Gaussian distributions, to construct the fitness function for classification. Two fitness measures, overlap area and weighted distribution distance, have been developed. Rather than using the best evolved program in a population, this approach uses multiple programs and a voting strategy to perform classification. The approach is examined on three multiclass object classification problems of increasing difficulty and compared with a basic GP approach. The results suggest that the new approach is more effective and more efficient than the basic GP approach. Although developed for object classification, this approach is expected to be able to be applied to other classification problems.

Previous work

[Zhan2003] Mengjie Zhang, Victor B.Ciesielski, and Peter Andreae, "A doman-independ window approach to multiclass object detection using genetic programming," *EURASIP Journal on Applied Signal Processing*, Vol.8, pp. 841-859, 2003.

Genetic programming(GP)

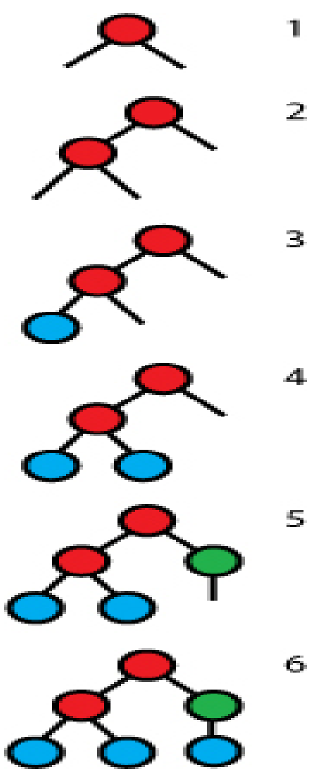
- Genetic programming is a branch of genetic algorithms.
- The main difference between genetic programming and genetic algorithms is the representation of the solution.



Initial population

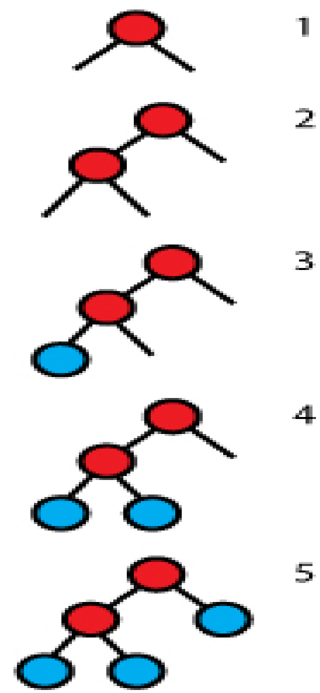
•We have determined the MAX DEPTH

1 、Full method



•The length of every path between a terminal and the root is equal to a predefined depth

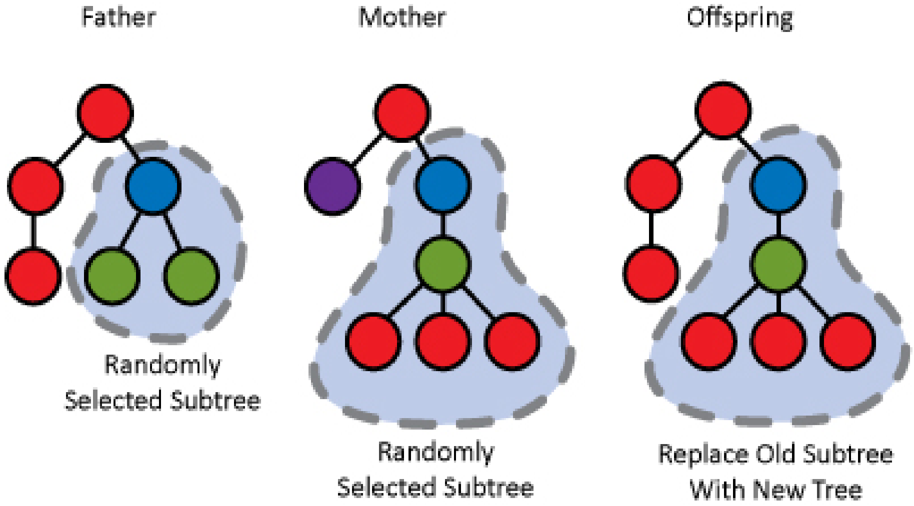
2 、Grow method



•The length of a path between a terminal and the root is at most a predefined maximum depth.

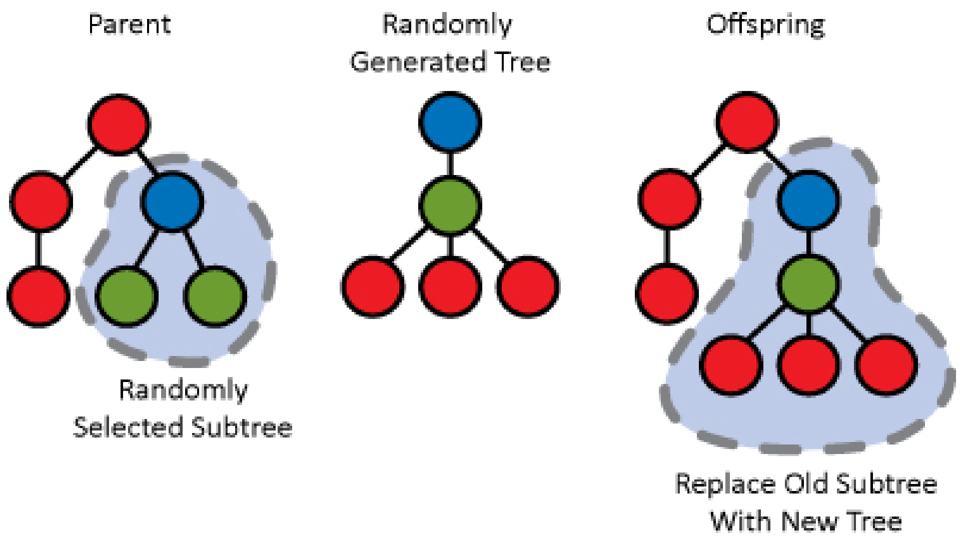
Usually, an equal mix of trees is produced with the Full and Grow methods.

Crossover(Subtree crossover)



The probability distribution for the selection of a crossover point favors the exchange of subtrees rather than terminals.

Mutation(Subtree mutation)



- First, a random point is chosen in the tree.
- This point, as well as the subtree below it, are then removed and replaced by a new, randomly generated subtree.

Terminals

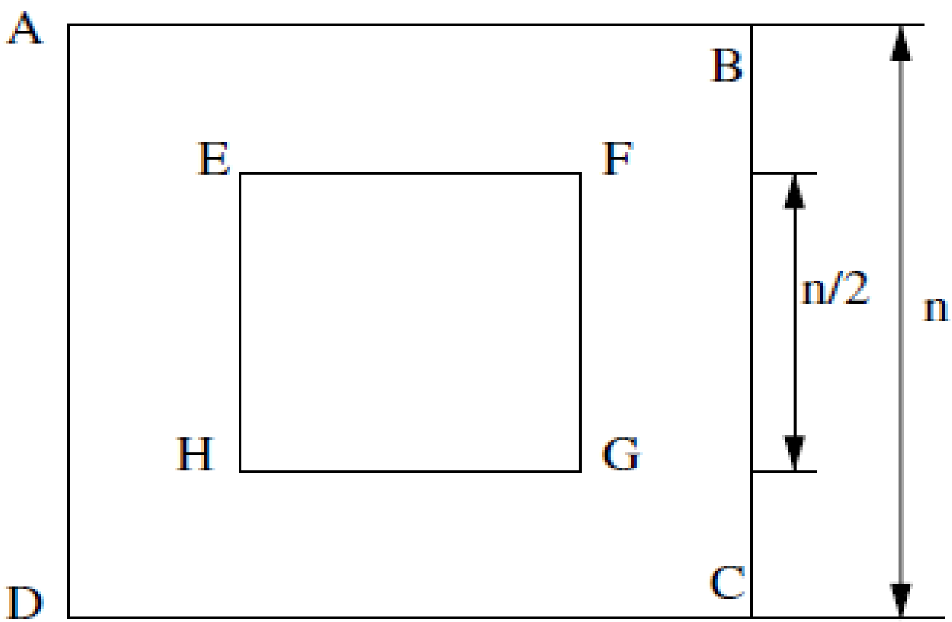


Fig. 1. Region features as terminals.

[Zhan2003] ABCD represents the input field, and EFGH represents the object.

Image features, pixel level, domain independent statistical features.

Here use the means and variances (pixels) of certain regions in object cutout images.

Also used some constants which generated from uniform distribution ([Zhan2003] range 0~255)) as terminals

Functions

Functions set = { +, - , *, /. if}

Fitness function

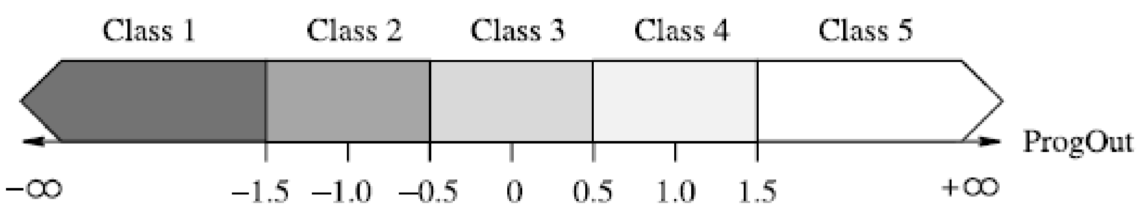


Fig. 2. An example program classification map.

$$\textbf{class} = \begin{cases} \text{Class } 1, & r < -1.5 \\ \text{Class } 2, & -1.5 \leqslant r < -0.5 \\ \text{Class } 3, & -0.5 \leqslant r < 0.5 \\ \text{Class } 4, & 0.5 \leqslant r < 1.5 \\ \text{Class } 5, & 1.5 \leqslant r \end{cases} \quad (2)$$

where r is the program output.

The *program classification map was* proposed in Zhan2003.

The boundary is defined by the user, and the ordering of class is fixed.

Fitness function (using Gaussian distribution)

Assume the behavior of a program classifier is modeled using multiple Gaussian distributions, each of which corresponds to a particular class.

Take two class problem as example.

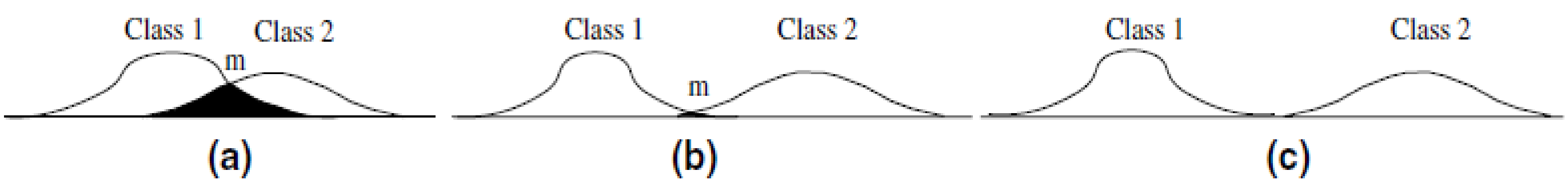


Fig. 3. Example normal distributions for a two-class problem.

Two measures: area and distance.

Area measure:

The overlap area of the two distributions.
(The smaller, the better)

$$P(x) = \frac{\exp(\frac{-x^2}{2})}{\sqrt{2\pi}}$$ (standard normal distribution)

$$A(x) = \sum_{i=0}^{x/\alpha} \alpha P(\alpha i)$$ (approximated of the area under the distribution at point)

$$A(\mu, \sigma, x) = A\left(\frac{x-\mu}{\sigma}\right)$$

The approximation of the overlap area A_o:

$$A_o = 1 - A(\mu_1, \sigma_1, m) - A(\mu_2, \sigma_2, m)$$

$0 \leq A_o \leq 1$

Distance measure:

The overlap of the two distributions.
(The smaller, the better)

$$d = 2 \times \frac{|\mu_1 - \mu_2|}{\sigma_1 + \sigma_2} \qquad d_s = \frac{1}{1+d} \qquad 0 \le d_s \le 1$$

Fitness function:

The number of classes is *n*. There are *C(n, 2)* overlaps of distributions.

$$\text{fitness} = \sum_{i=1}^{C_n^2} M_i$$

M_i is a fitness measure of the *i*th overlap of distribution of two classes, which can be either the <u>area measure</u> or the <u>distance measure.</u>

Obtaining classification accuracy

The probability $prob_c$ of a given pattern being of class c.

$$Prob_c = \prod_{i=1}^{I} P(\mu_{i,c}, \sigma_{i,c}, r_i) \qquad P(\mu, \sigma, x) = \frac{\exp\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}{\sigma\sqrt{2\pi}}$$

Where

- $P(.)$ is the Gaussian p.d.f.,
- r_i is the output result of program i with the pattern to be classified,
- I are the programs in the population,
- $\mu_{i,c}$ and $\sigma_{i,c}$ are the mean and standard deviation of the outputs of program i for class c.

Note： the formula is correct only when the output of the different classifiers/programs are independent random variable, and it will be improved in the future work.

Experiment

Data set:

1.Shares (3 classes, 960 objects)

The pixels of the objects were produced using a Gaussian generator with different means and variances for each class.

2. Coins (5 classes, 576 objects)

Scanned 5 cent and 10 cent New Zealand coins which were located in different sides (head and tail). The background was quite cluttered.

3. Faces (4 people, 40 objects)

Human faces taken at different times, varying lighting slightly, with different expressions. Images are collected from ORL face database.

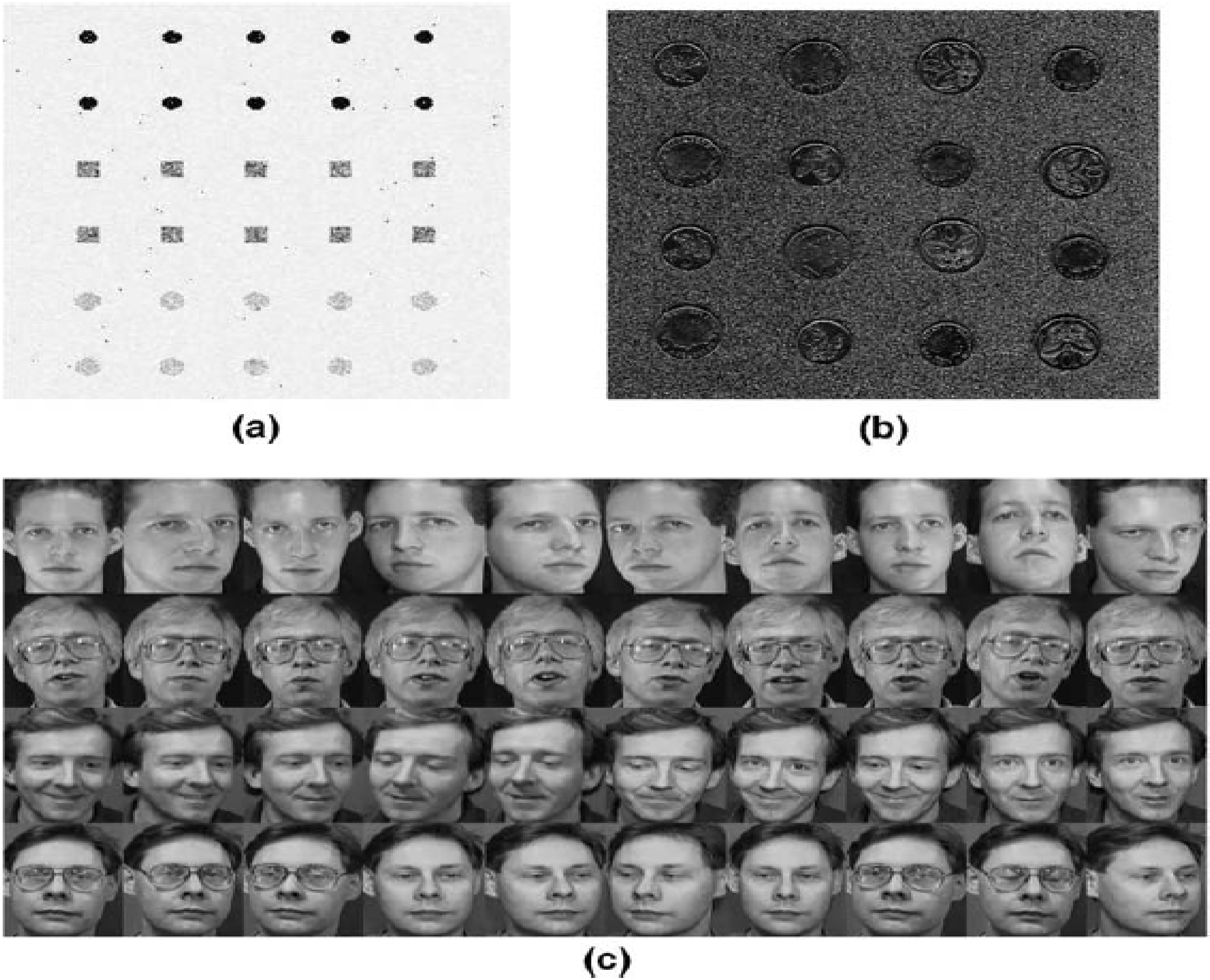


Fig. 4. Data set examples: (a) shapes, (b) coins, and (c) faces.

Table 1
Parameters used for GP training for the three data sets

Parameter names	Shapes	Coins	Faces
Population-size	300	500	500
Initial-max-depth	3	3	3
Max-depth	5	8	8
Max-generations	51	51	51
Object-size	16×16	70×70	92×112

Parameter names	Shapes (%)	Coins (%)	Faces (%)
Reproduction-rate	10	10	10
Cross-rate	60	60	60
Mutation-rate	30	30	30
Cross-term	15	15	15
Cross-func	85	85	85

- Average best classification results (over 500 experiments).
- Both methods use the same parameter.
- Single best evolved program is used in basic GP.
- Multiple best evolved programs and a simple vote from these programs are used in new approach.

Table 2
Best results of the new and the basic GP approaches

Data set	Strategy	Generations	Time (s)	Test accuracy (%)
Shapes	Basic approach	11.70	3.29	98.64
	New approach	0.86	0.29	99.96
Coins	Basic approach	41.02	6.07	90.46
	New approach	14.26	2.27	98.60
Faces	Basic approach	8.60	0.38	86.75
	New approach	4.66	0.24	97.75

Table 3
Results for different programs and the two fitness measures

Data set	Programs used	Fitness measures					
		Generations		Time (s)		Test accuracy (%)	
		Distance	Area	Distance	Area	Distance	Area
Shapes	1	0.88	0.86	0.26	0.29	97.48	99.96
	3	0.38	0.34	0.19	0.20	99.84	99.83
	5	0.14	0.12	0.15	0.16	99.81	99.80
	10	0.04	0.04	0.14	0.15	99.79	99.78
	20	0.06	0.06	0.16	0.18	99.68	99.68
	50	0.30	0.16	0.30	0.27	99.46	99.46
Coins	1	32.98	33.78	4.55	5.37	94.23	97.49
	3	18.70	19.42	2.47	2.93	97.81	98.41
	5	14.96	15.76	1.96	2.39	98.12	98.38
	10	14.62	14.26	2.06	2.27	98.46	98.60
	20	10.82	9.06	1.60	1.57	98.40	98.25
	50	14.54	13.32	2.84	2.90	98.06	98.29
Faces	1	5.13	4.66	0.20	0.24	96.35	97.75
	3	2.09	1.73	0.08	0.10	96.95	95.90
	5	1.56	1.49	0.07	0.09	96.10	96.10
	10	1.03	0.88	0.05	0.06	95.45	94.70
	20	0.64	0.68	0.04	0.06	93.25	93.25
	50	0.25	0.18	0.03	0.04	91.20	90.30

Thank you!!